

Final-round design shortlist

floodlight · round 11 · aerodynamics-r3-cb659d5e, structures-r2-c44c2a7e, weights-r2-3ba60b4e

20 admissible designs drawn from the final campaign round. 400,000 points were sampled inside the round-11 trust region; 18.26% of them are **admissible** — all six disciplinary margins satisfied *and* all three coupling assumptions self-consistent. These are the lightest members of separated neighbourhoods within that set, so they are 20 genuinely different aircraft rather than 20 samples of one.

What backs these numbers

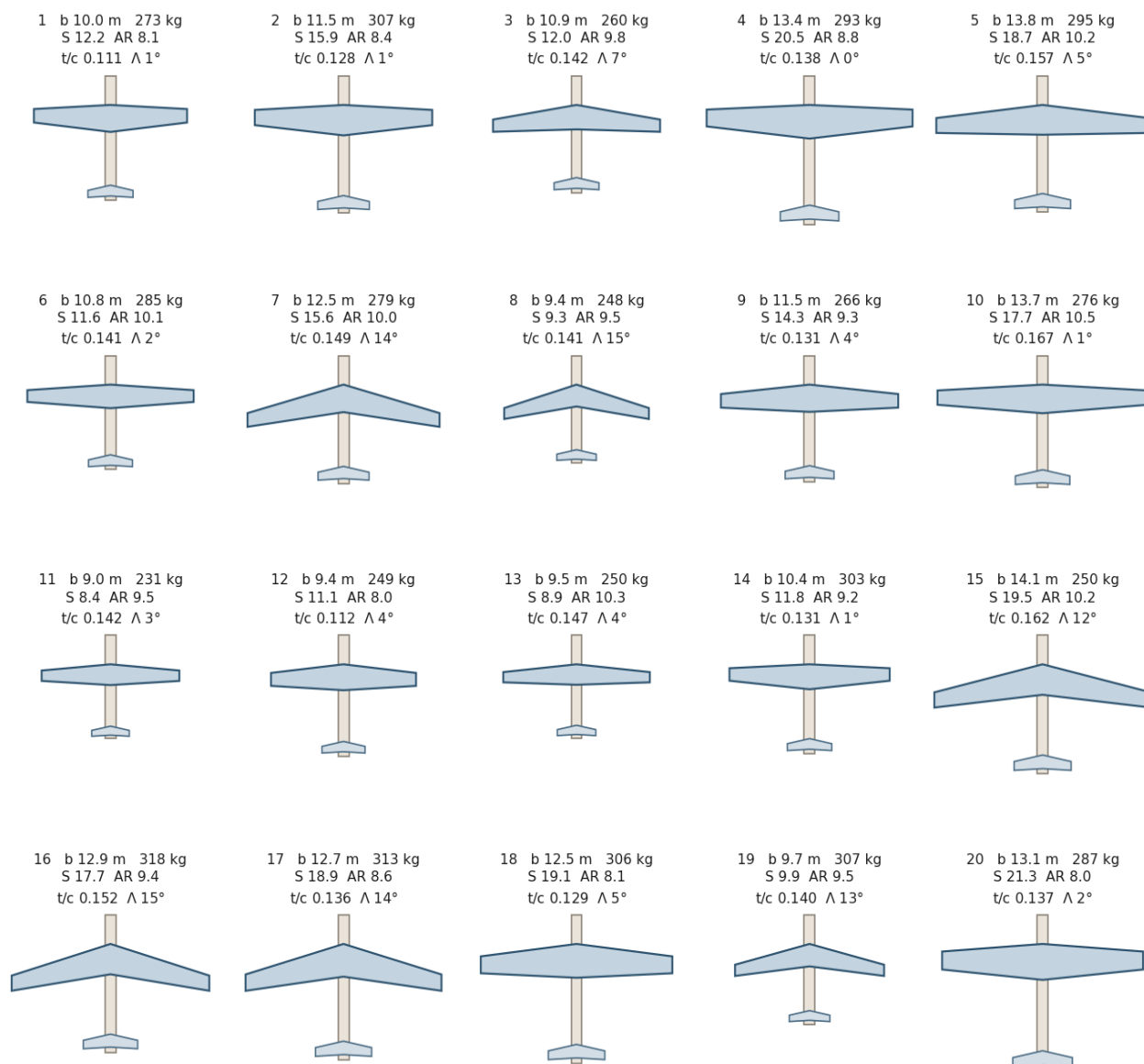
Aerodynamics is an AVL vortex-lattice solve coupled strip-by-strip to NeuralFoil section polars, verified against the Warren 12 planform to 0.14% on lift-curve slope. Structures is a fully-stressed wing-box beam sized against the AVL spanload, with cap area scaled up wherever stiffness rather than strength governs. Weights is a component build-up: a fuselage sized by the volume it must hold and weighed by wetted area, plus propulsion from installed power. No coefficient in this document was hand-written.

The 20 designs

#	S_ref m ²	AR	t/c	sweep deg	W0 kg	P_SL kW	span m	W/S kg/m ²	W_empty kg	W_str kg
1	12.19	8.14	0.111	0.5	500	88	9.96	41.0	273	31
2	15.95	8.35	0.128	0.7	500	115	11.54	31.4	307	29
3	12.04	9.81	0.142	7.3	500	80	10.87	41.5	260	31
4	20.45	8.77	0.138	0.2	500	93	13.39	24.4	293	31
5	18.68	10.24	0.157	4.7	500	103	13.83	26.8	295	33
6	11.57	10.11	0.141	1.7	500	107	10.82	43.2	285	32
7	15.57	10.02	0.149	13.6	500	92	12.49	32.1	279	33
8	9.34	9.48	0.141	14.9	500	74	9.41	53.6	248	27
9	14.25	9.33	0.131	3.6	500	80	11.53	35.1	266	33
10	17.75	10.53	0.167	1.2	500	87	13.67	28.2	276	31
11	8.44	9.50	0.142	2.8	500	58	8.95	59.2	231	25
12	11.09	8.02	0.112	3.9	500	66	9.43	45.1	249	29
13	8.86	10.26	0.147	3.8	500	78	9.53	56.5	250	28
14	11.81	9.20	0.131	0.5	500	123	10.42	42.4	303	30
15	19.47	10.18	0.162	11.8	500	56	14.08	25.7	250	32
16	17.66	9.35	0.152	14.7	500	126	12.85	28.3	318	29
17	18.89	8.60	0.136	14.1	500	116	12.74	26.5	313	30
18	19.08	8.13	0.129	4.5	500	107	12.46	26.2	306	29
19	9.85	9.54	0.140	12.6	500	133	9.69	50.8	307	28
20	21.28	8.02	0.137	1.6	500	82	13.07	23.5	287	27

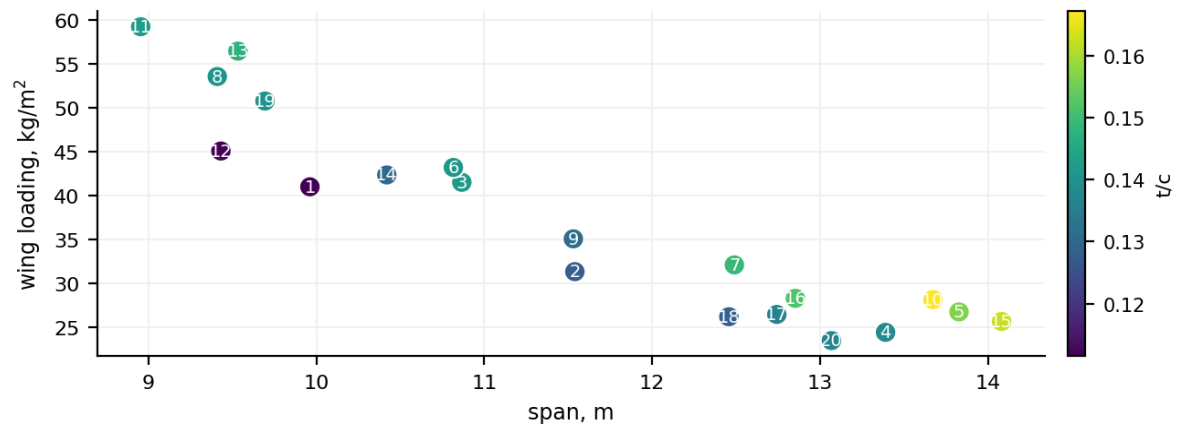
W_empty and W_str are produced by the weights and structures publications at each design point, not assumed.

The aircraft, to scale



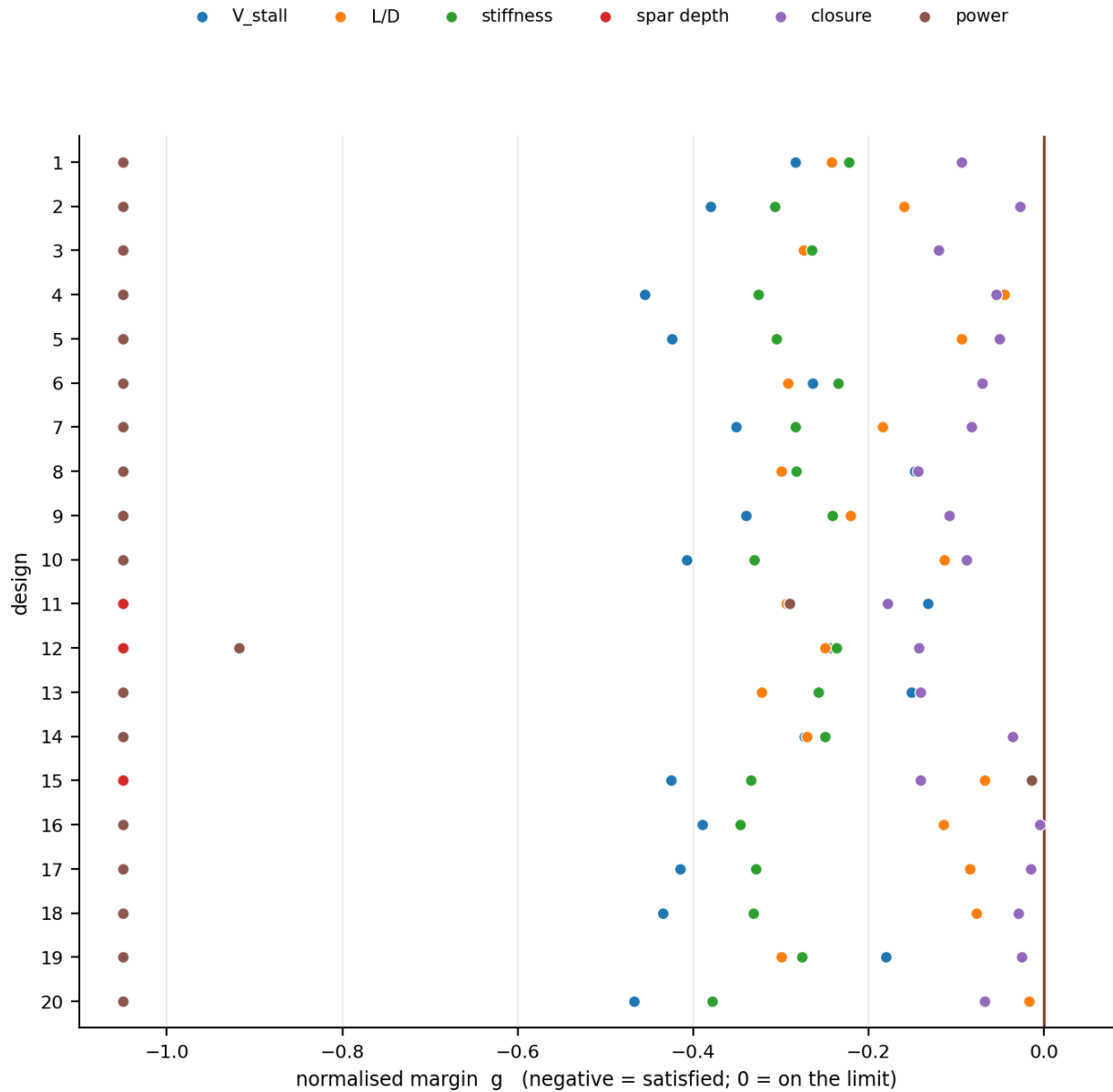
Top views to a single common scale, so spans compare directly. Wing planform and tail from the shared configuration (taper 0.5, tail volume 0.55); fuselage from the weights build-up, whose length now obeys the config rule $\max(\text{volume-driven}, \text{tail-arm-driven})$ — every body here is long enough to carry its own tail, and the dashboard draws from the same geometry. These are the dimensions the publications were built on.

How they differ



Span runs from 9.0 m to 14.1 m at essentially the same gross mass. That spread is the point: the admissible set is a region, and a single 'optimum' would misrepresent it.

Margins



Every constraint is normalised so that $g \leq 0$ means satisfied and the value is the fraction of the limit remaining. A structures margin of -0.10 and an aero margin of -0.10 mean the same thing.

Read this before using any of it

These are not optima. Only 9 constraints are active across the shortlist. A true optimum sits on a corner with several constraints simultaneously active. These do not, because gross mass is bounded below at 500 kg by the sampling region — structures' declared validity floor — not by physics: the lightest design here is 500 kg, and it is light because the search box stops there. The real minimum is outside the region the campaign converged on.

Admissibility now means "where the teams' beliefs hold". Consistency is checked against each discipline's LOSSY internalization of its partners — a deliberately low-order fit, the way a receiving team actually holds an expert's model. These designs sit where that simplified picture is valid: W_{empty} residuals run 0.01–0.42 of tolerance here. Designs outside the shortlist are often rejected not because the aircraft fails, but because the teams' current understanding of each other cannot vouch for that corner of the space. That is the intended meaning, and it is worth keeping distinct from physical infeasibility.

The endurance requirement is not enforced anywhere. The reference mission asks for 8 hours at 40 m/s. No constraint in any publication checks it; cruise L/D stands in as a proxy and fuel fraction is a fixed 18% constant rather than an outcome. No propulsion discipline exists yet, so nothing owns P_{SL} or closes the range equation.

The planform is pinned, not designed. Taper 0.5, -2° washout, NACA 24xx sections, no fuselage or tail in the aerodynamic model. Thickness is quantised to whole percent by the 4-digit section. These are assumptions shared by all three disciplines and they are not design variables.

Structures ignores buckling. Caps are sized on stress alone, which is optimistic for the compression cap. The validity box is clipped to where the cap area stays under $3\times$ strength-sized, but within that region the model still has no torsion, no aeroelastic feedback, no gust cases and no landing loads.

The fuselage/tail mismatch was caught and fixed. Earlier in this project, weights sized the body on internal volume alone while aerodynamics assumed a tail mounted 2.9–5.0 m aft of where that body ended — roughly 43 kg of structure nobody was paying for, invisible to every consistency check because the interface was never declared. The shared configuration now sizes every fuselage to $\max(\text{volume-driven}, \text{tail-arm-driven})$ length, weights carries the mass, and the drawings above use the same rule. It is listed here because the class of error — an undeclared interface between teams — is not fixable in general, only findable.

Non-wing drag is unresolved. The aerodynamic publication carries a declared parasite-drag constant for the fuselage and tail. A wetted-area build-up from the weights model gives 25–37% less. The two have not been reconciled, and the aero publication has deliberately not been switched over to the lower number.

Conditioning at round 11: $W_{\text{empty}} = 354.5$, $W_{\text{structure}} = 39.4$, $CL_{\text{max}} = 1.6$. Every design above satisfies all three within the tolerances the publications declare.